LYLE FORSYTH ASSOCIATES
Consulting Building Services & Façade Engineers
Pavilion House, 9 Newhall Street, Birmingham B3 3PJ

INVESTIGATION REPORT

The Vault Birmingham, 118 Broad Street, Birmingham B15 1AU
Defects investigation — pool hall envelope, water-treatment and ventilation

Client: Brampton Leisure Limited **Ref:** LFA-2024-318 **Date:** 18 October 2024

Prepared by: Dr Margaret Lyle CEng MCIBSE & Andrew Forsyth CEng MIStructE

1. Instructions and scope

1.1 We were instructed by Brampton Leisure Limited, through Whitfield Carter LLP, to investigate a series of defects affecting the pool hall at The Vault Birmingham and to express opinions on their cause. Our investigation comprised four site visits between 24 September and 11 October 2024, airflow and humidity measurement, inspection of the water-treatment plant and commissioning records, and review of the design information and the Employer's Requirements.

2. The water ingress

2.1 Persistent water ingress was observed at the junction of the curtain-wall glazing and the roof abutment along the south and west elevations of the pool hall. Staining and active corrosion were evident on the exposed structural steelwork.

2.2 On opening up, we found that the abutment flashing and the curtain-wall head seals had been installed without a continuous secondary line of defence, contrary to good practice and to the CWCT Standard. The detailing did not accord with the weather-tightness performance required by the Employer's Requirements. In our opinion this is a defect of both design (within the Contractor's Designed Portion) and workmanship.

3. The pool water-treatment plant

3.1 The Employer's Requirements (Section M3) require a turnover period consistent with PWTAG guidance for a competition-class 25-metre pool at the stated bathing load. The installed filtration plant provides a turnover materially longer than required; in our calculation the plant is undersized by approximately 18% against the M3 criterion.

3.2 The commissioning records are incomplete. The dosing control was, in our opinion, not correctly commissioned, which is consistent with the documented combined-chlorine and pH excursions and the three closures directed by the Environmental Health Officer.

4. The pool hall ventilation

4.1 We measured the supply airflow to the pool hall and found it to be approximately 22% below the design rate necessary to control humidity within the parameters of Employer's Requirements Section M4 and CIBSE TM58. With the pool operated within the design water

and air temperatures, the installation cannot maintain the design relative humidity. This has caused the observed condensation, mould growth and deterioration of finishes.

4.2 We do not accept that the condensation is attributable to operation outside the design basis: our measurements were taken with the pool operating within the design envelope.

5. Studio HVAC

5.1 The controls serving Studios 2 and 3 were incorrectly configured and the zoning did not match the design. This is a workmanship/commissioning defect.

6. Conclusions

6.1 In our opinion the water ingress, the undersized and incorrectly commissioned water-treatment plant, the undersized ventilation and the studio HVAC faults are each the result of design and/or workmanship failings on the part of the Contractor and of non-compliance with the Employer’s Requirements. The estimated cost of the remedial works necessary to put the defects right, prepared with the project quantity surveyor, is approximately £1.757 million exclusive of VAT.

Dr Margaret Lyle CEng MCIBSE | **Andrew Forsyth** CEng MIStructE

Lyle Forsyth Associates — 18 October 2024